

Name: _____

Date: _____



Year 8 Mathematics 2015

Test 2: General and Extension

Topics – Number and Algebra

 70

= _____ %

Total Time: **60 minutes**
 Total Reading: **5 minutes**
 Total Working: **55 minutes**
 Weighting: **% of the year.**

Bonus Mark

☐ Notes Page (1 Mark)

CALCULATOR FREE

Time:	<i>42 minutes</i>	Marks for Section 1:	<i>43</i>
Reading:	<i>2 minutes</i>	Equipment Allowed:	<i>1 A4 page of notes</i>
Working:	<i>40 minutes</i>		

1.	[20 Mark] 1,1,1,1,1,1,1,1,1,1,1,1,1,2,2		
	Calculate:		
a	$6 + 32 =$ 38 ✓	b	$12 - 4 =$ 8 ✓
c	$12 \times 6 =$ 72 ✓	d	$54 \div 6 =$ 9 ✓
e	$30.6 + 7.5 =$ 38.1 ✓	f	$17 - 22 =$ -5 ✓

g	$-4 \times 8 =$ -32 ✓	h	$-21 \div -3 =$ 7 ✓
i	$1376 + 46 + 321 =$ $\begin{array}{r} 1376 \\ 46 \\ 321 \\ \hline 1743 \end{array}$ ✓	j	$86.8 - 1.3 =$ 85.5 ✓
k	$4 \times -12 =$ -48 ✓	l	$96 \div 8 =$ 12 ✓
m	$-12 + 28 =$ 16 ✓	n	$-12 + (-17) =$ -29 ✓
o	$(-15) \times (-2) =$ 30	p	$5 \times (-6) =$ -30 ✓
q	$(2+7) \times 9 =$ 9 ✓ 81 ✓	r	$7 - 16 + (21 \div 3) \times (5 - 2) =$ $7 - 16 + 21 \div 3 \times 3 =$ 12 ✓

2.	[5 marks] Use the following words from the list below to complete the following sentences. (Hint: There are more words than needed) Base, Expanding, Rational, Number, Algebraic, Factorising, Simplifying, Index, one.		
a)	<u>Expanding</u> is when you remove brackets and <u>factorising</u> is when you replace the brackets. ✓		
b)	A power is written with the <u>base</u> number at normal size and a small number at the top right called the <u>index</u>. ✓		
c)	Any number raised to the power of zero equals <u>one</u>. ✓		
3.	[1 marks] Using all the digits 8, 9, 1, 5, 6 (once only) to create a number that is closest to 90. An example would be 98.156 89651 ✓		
4	[4 marks; 1,1,1,1] Round the following decimals to 2 decimal points. (2 d.p)		
a	26.3245	b	2.93933
	26.32 ✓		2.94 ✓
c	0.8253	d	4.76439211
	0.83 ✓		4.76

5.	[5 marks: 1, 1, 1, 2] Write the following in index form.		
a	$3 \times 3 \times 3 =$ 3^3 ✓	b	$5 \times 5 \times 5 \times 5 \times 5 =$ 5^5 ✓
c	$6 \times 6 =$ 6^2 ✓	d	$4 \times 4 \times 9 \times 9 \times 9 =$ $4^2 \times 9^3$ ✓✓
6.	[8 marks, 2,2,2,2) Use the index laws to find the following as simple powers.		
a	$4^2 \times 4^3 =$ 4^5 ✓✓	b	$7^4 \times 7^8 =$ 7^{12} //
c	$5^7 \div 5^2 =$ 5^5 //	d	$6^{23} \div 6^{12} =$ 6^{11} //

END OF SECTION ONE

Name: _____

CALCULATOR ALLOWED

Time:	<i>20 minutes</i>	Marks for Section 2:	<i>27</i>
Reading:	<i>2 minutes</i>	Equipment Allowed:	<i>1 A4 page of notes</i> <i>Calculator</i>
Working:	<i>22 minutes</i>		

7. [7 marks; 1,1,2,3]

Simplify the following expressions;

a) $7m + 4m =$

$11m$ ✓

b) $x - 2 + 4x + 8 =$

$5x + 6$ ✓

c) $t + 4s - 2t + s + 6t =$

$5t + 5s$ ✓✓

d) $h + z + 6 - 3z + 7h + 9 - 2z + 3h - 1 =$

$11h - 4z + 14$ ✓✓✓

8.	[4 marks; 1,1,2]	
	Expand each of the following;	
a	$6(z + 9) =$ $6z + 54$ ✓	b $-2(t + 7) =$ $-2t - 14$ ✓
c	$4(w + 3) + 7(2w - 8) =$ $4w + 12 + 14w - 56$ $18w - 44$ ✓✓	
9.	[5 Mark 1, 2, 2]	
	Calculate the value of the statement: when $x = 3$	
a	$3x + 5 =$ $9 + 5$ $= 14$ ✓	b $x(2 + 5) =$ $3(7)$ $= 21$ ✓✓
c	$\frac{18 - x}{3} =$ $\frac{18 - 3}{3} = \frac{15}{3} = 5$ ✓✓	
10.	[4 marks; 1,1,2]	
	Calculate the value of the statement: when $x = -5$	
a	$2x$ -10 ✓	b $4x + 8 =$ $-20 + 8$ -12 ✓
c	$2(x + 9) =$ $2(-5 + 9)$ $= 2(4)$ $= 8$ ✓✓	

10.	[7 marks; 2,2,3]		
	Factorise each of the following:		
a	$6x + 30 =$ $6(x+5)$ ✓✓	b	$35a - 20c =$ $5(7a - 4c)$ ✓✓
c	$35a + 15b - 20c =$ $5(7a + 3b - 4c)$ ///		

END OF SECTION TWO